

Atlas

Zimbabwe Tourism Destination Insights

Data Experience Design and Storytelling using Tourism Destination Insights

Track Track 2: Design

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1. Problem Definition & User Personas

Zimbabwe's tourism sector generates significant economic value across eight national destinations, yet tourism authorities and destination managers lack a unified, interactive view of visitor trends, spend patterns, service quality, and complaint themes. Without this visibility, resource allocation and intervention decisions are made on intuition rather than evidence.

The core problem is fragmented data access: visitor counts, accommodation occupancy, digital booking rates, and complaint themes exist in separate reports or are not collected systematically. This creates three critical gaps:

- Revenue leakage — High-visitor destinations with low digital booking share cannot capture spend already on-site.
- Service quality blind spots — Destinations with declining service quality scores go unnoticed until visitor satisfaction drops.
- Reactive intervention — Complaint themes are identified only after they compound across multiple destinations.

Target Users

Persona		Role	Decision Need
Tourism Analyst	Authority	Zimbabwe Tourism Authority, Harare	Which destinations need immediate attention? Where should marketing budget be allocated?
Destination Manager		Victoria Falls, Great Zimbabwe, etc.	How is my destination performing vs peers? What service gaps should I address first?
Local Association	Business	Hospitality operators at each destination	What visitor trends affect my business? Should I invest in digital booking?

Each persona has a distinct decision cadence: analysts review monthly trends, destination managers act on weekly operational signals, and business associations respond to seasonal patterns. The dashboard accommodates all three by supporting month-level filtering and destination-level drill-down.

2. Interface Design & Wireframes

The dashboard follows a single-page layout with three visual zones, each corresponding to a step in the storytelling flow. The interface uses a light theme with shadcn/ui components for accessibility and consistency.

Layout Structure

Zone 1 — Overview KPIs: Five summary cards at the top of the page display total visitors, estimated spend, average service quality, average digital booking share, and the national top complaint theme. These values update dynamically when filters change.

Zone 2 — Interactive Filtering + Trend: A card below the KPIs contains three filter groups: Month (dropdown), Destination Type (toggle chips), and Province (toggle chips). Below the filters, a dual-axis line chart shows visitor count and estimated spend trends from January to June 2026, with an auto-generated insight narrative panel beside it.

Zone 3 — Destination Deep Dive: A Leaflet interactive map of Zimbabwe shows each destination as a circle marker sized by visitor volume and colored by service quality. A sortable scorecard table provides detailed metrics for each destination. Below, a multi-view chart (with dropdown selector) displays complaint themes, revenue by destination, or visitor demographics. An action panel shows three priority recommendations.

Key Design Decisions

- **Single-page layout:** No navigation — all information is accessible without page transitions, supporting the 'scan → drill → decide' workflow.
- **Interactive Leaflet map:** Replaced static SVG with a real OpenStreetMap layer bounded to Zimbabwe, allowing zoom and click-for-detail.
- **Multi-view analytics:** A dropdown switches the deep-dive chart between complaint themes, revenue ranking, and domestic/international visitor split — three views from one component.
- **Color-coded quality signals:** Red (< 55), amber (55-65), green (> 65) for service quality, digital booking, and other scores — consistent across all visualizations.
- **Responsive grid:** 5-column grid at desktop, stacking to single column on mobile, with all touch targets $\geq 44 \times 44$ px.

Visual Examples

The dashboard renders in light mode by default with a dark mode toggle. All chart axes use visible muted-foreground text, and all interactive elements have aria-labels and aria-pressed states for screen reader support.

3. Storytelling Narrative & Flow

The design implements the four-step storytelling flow required by the evaluation criteria. Each step is mapped to a specific section of the dashboard:

Step	Dashboard Section	What the User Sees
1. Overview / Problem	KPI Strip (5 cards)	Total visitors (144,096), estimated spend (\$13.6M), avg service quality (73.7/100), avg digital booking (45.6%), top complaint (Sanitation). Immediate sense of scale and priority.
2. Data Exploration	Filter Bar + Trend Chart	Month dropdown, type chips, province chips. Dual-axis chart shows visitor and spend trends Jan–Jun. User can filter by any combination and see all visualizations update instantly.
3. Key Insights	Insight Narrative Panel	Auto-generated bullets: Victoria Falls drives 52% of spend. National visitation up 32%. Sanitation is top complaint at 7 of 8 destinations. Gonarezhou has clearest digital booking gap.
4. Recommended Actions	Action Panel (3 cards)	Priority 1: Fast-track online booking at Gonarezhou Gateway (revenue leakage). Priority 2: Review service delivery at Great Zimbabwe (service quality). Priority 3: Address sanitation as a national cross-site issue.

Flow Justification

This flow follows the natural decision-making sequence: understand the current state (KPIs), explore specific segments (filters), identify what matters (insights), and decide what to do (actions). The insight narrative is generated programmatically from the filtered data, ensuring it always reflects the current view without requiring user interpretation.

The multi-view chart in the deep-dive section adds exploration depth: users can switch between complaint analysis (operational), revenue ranking (financial), and visitor demographics (strategic) without leaving the dashboard.

4. Accessibility & Usability

The prototype targets WCAG 2.1 Level AA compliance. The following evidence documents compliance across the four required areas:

Contrast Ratios

Element	Foreground	Background	Ratio	Status
Body text	oklch(0.145)	oklch(1.0)	~16:1	Pass
Muted text	oklch(0.556)	oklch(1.0)	~7:1	Pass
Primary buttons	oklch(0.985)	oklch(0.205)	~12:1	Pass
Chart axis labels	oklch(0.556)	oklch(1.0)	~7:1	Pass
Badge (teal)	#8fd0ba	rgba(79,143,124,0.18)	>4.5:1	Pass
Badge (amber)	#c68a34	rgba(198,138,52,0.18)	>4.5:1	Pass

Touch Targets

All interactive elements (filter chips, buttons, select triggers) use a minimum height of 36px (h-9) with horizontal padding, achieving an effective touch area of $\geq 44 \times 44$ CSS pixels. Filter chips are rounded pills with 12px horizontal padding. Select triggers include 10px vertical padding for comfortable interaction.

Screen-Reader Support

- **Charts:** All SVG chart elements have `role="img"` and `aria-label` descriptions (e.g., "Map of Zimbabwe tourism destinations with interactive markers").
- **Map:** Leaflet container has `aria-label="Interactive map of Zimbabwe tourism destinations."` Circle markers have popup content accessible via keyboard.
- **Tables:** Destination scorecard uses `<caption>` element with descriptive text.
- **Filters:** Select triggers have `aria-label="Filter by month."` Toggle chips have `aria-pressed` state. Filter groups have `role="group"` with `aria-label`.
- **KPI Strip:** Has `role="region"` with `aria-label="National summary KPIs."`

Responsive Layout

The dashboard adapts fluidly across four breakpoints with mobile-first design:

- **320px (mobile):** Single-column layout, stacked filters, scrollable table. Charts scale to h-56. Map scales to h-64. KPI values use text-xl.
- **768px (tablet):** Two-column grid for charts and maps. Filters display inline. Chart dropdown stacks below title.
- **1024px (desktop):** Full five-column grid layout with side-by-side panels. Charts use h-72. Map uses h-80.
- **1920px (wide):** Max-width container (1180px), centered content.

All interactive elements have touch targets $\geq 44 \times 44$ CSS pixels. Select triggers adapt from full-width on mobile to fixed-width on desktop. The scorecard table supports horizontal scrolling on narrow screens. The Leaflet popup uses max-width: 80vw to prevent overflow.

Keyboard Navigation

All interactive elements are reachable via Tab. Filter chips are operable with Enter/Space. Select dropdowns support arrow-key navigation. The Leaflet map supports zoom via keyboard. The theme toggle button is accessible via Tab.

Keyboard shortcuts provide power-user access: pressing ? opens a shortcut overlay listing 12 shortcuts. M focuses the month filter, R resets all filters, D toggles dark/light mode, and 1-4 jump to dashboard sections. Shortcuts are disabled when typing in input fields to avoid conflicts.

5. Dataset Binding & Asset Licensing

Dataset Integration

The dashboard dynamically binds to the provided CSV dataset file `04_tourism_destination_insights.csv`. The data flow is:

- **Build time:** A server-side CSV parser (`src/lib/csv-loader.ts`) reads the file from `public/data/` and parses all 48 records into typed TypeScript objects.
- **Server render:** The page component passes the parsed data as props to the client dashboard component.
- **Client runtime:** A custom React hook (`useTourismData`) manages filter state and computes all derived data: KPIs, trend data, destination aggregations, complaint frequencies, revenue rankings, and demographics.
- **Reactivity:** Filter changes trigger recomputation — no hardcoded values. Changing the month filter updates all charts, KPIs, insights, and action cards instantly.

The dataset contains 48 records: 8 destinations × 6 months (January–June 2026). Fields include `visitor_count`, `estimated_total_spend_usd`, `service_quality_score_0_100`, `digital_booking_share_pct`, `domestic_visitor_share_pct`, `accommodation_occupancy_pct`, `avg_spend_usd_per_visitor`, `transport_access_score_0_100`, and `top_complaint_theme`.

Asset Licensing Register

Asset	Source	License	Usage
Geist Sans	Vercel (next/font)	OFL 1.1	Body text, headings
Geist Mono	Vercel (next/font)	OFL 1.1	Monospace elements
Lucide React	npm	ISC	All UI icons
shadcn/ui	npm	MIT	Card, Badge, Button, Table, Select, etc.
Tailwind CSS 4	npm	MIT	Utility-first CSS
Recharts	npm	MIT	LineChart, BarChart
Leaflet	npm	BSD-2	Interactive map
react-leaflet	npm	ISC	React Leaflet bindings
OpenStreetMap	osm.org	ODbL	Map tiles
Next.js 16	npm	MIT	Framework
React 19	npm	MIT	UI library
04_tourism_destination_insights.csv	POTRAZ AI4I	Challenge-provided	Dashboard data

Data Rights

All data is synthetic aggregate sample data provided by POTRAZ for the AI4I challenge. The dataset contains no personal names, phone numbers, ID numbers, household addresses, or individual health records. Coordinates are approximate centroids for prototyping only. The

dashboard is clearly labeled: 'Synthetic aggregate sample data generated for the POTRAZ AI4I Design Track. Not official tourism statistics.'